

For the use of a Registered Medical Practitioner or a Hospital or a Laboratory

Alfacip 1 mcg

(Alfacalcidol capsules 1 mcg)

Composition

Each soft gelatin capsule contains

Alfacalcidol BP1 mcg

(1 -Hydroxyvitamin D3)

Approved colours used in capsule shell

Ponceu 4R BIS 16255

Description:

Red transparent round shaped soft gelatin capsules.

Pharmacology

Pharmacodynamics

Alfacalcidol undergoes rapid hepatic conversion to 1,25-dihydroxyvitamin D3 metabolite which acts as a regulator of calcium and phosphate metabolism which acts as a regulator of calcium and phosphate metabolism. The main effects are to increase circulating 1,25-dihydroxyvitamin D3 levels and thereby to increase intestinal absorption of calcium and phosphate, promote bone mineralization, decrease plasma parathyroid hormone levels as well as to decrease bone resorption, with relief of bone and muscle pain. Treatment of intestinal calcium malabsorption associated with postmenopausal, senile and steroid induced osteoporosis, with alfacalcidol improves the negative calcium balance of these patients.

The more positive calcium balance, produced by alfacalcidol treatment in osteoporotic patients,decreases bone loss and bone fracture.

Pharmacokinetics

In patients with renal failure, 1-5 µg/day of 1 -hydroxyvitamin D (1 -OHD3) increased intestinal calcium and phosphorus absorption in a dose-related manner. This effect was observed within 3 days of starting the drug and conversely, it was reversed within 3 days of its discontinuation.

In patients with nutritional osteomalacia, increases in calcium absorption were noted within 6 hours of giving 1 µg 1 -OHD3 orally and usually peaked at 24 hours. 1 -OHD3 also produced increases in plasma inorganic phosphorus due to increased intestinal absorption and renal tubular re-absorption. This latter effect is a result of PTH suppression by 1 -OHD3. The effect of the drug on calcium was about double its effect on phosphorus absorption.

Patients with chronic renal failure have shown increased serum calcium levels within 5 days of receiving 1 -OHD3 in a dose of 0.5 - 1.0 µg/day. As serum calcium rose, PTH levels and alkaline phosphatase decreased toward normal.

Indication

Alfacalcidol is used for treating conditions in which calcium metabolism is disturbed due to impaired 1-alpha hydroxylation, such as reduced renal function; in other disorders associated with Vitamin D resistance; and in calcium malabsorption of osteoporosis.

The main indications are:

1. Renal bone disease (renal osteodystrophy)
2. Hypoparathyroidism
3. Hypoparathyroidism (with bone disease)
4. Nutritional and malabsorptive rickets and osteomalacia
5. Hypophosphataemic Vitamin D-resistant rickets and osteomalacia
6. Pseudo-deficiency (D-dependent Type I) rickets with osteomalacia
7. Osteoporosis

Dosage & Administration

Route of administration: oral

Initial dose for all indications:

Adults	1 microgram/day
Dosage in the elderly	0.5 microgram/day
Neonates and premature infants	0.05 - 0.1 microgram/kg/day
Children under 20 kg bodyweight	0.05 microgram/kg/day
Children over 20 kg bodyweight	1 microgram/day

The dose of Alfacalcidol should be adjusted thereafter to avoid hypercalcaemia according to the biochemical response. Indices of response include plasma levels of calcium (ideally corrected forprotein binding), alkaline phosphatase, parathyroid hormone, as well as radiographic and histological investigations. Plasma levels should initially be measured at weekly intervals. The daily dose of Alfacalcidol may be increased by increments of 0.25 - 0.5 microgram. When the dose is stabilised, measurements may be taken every 2 - 4 weeks.

Most adult patients respond to doses between 1 and 3 micrograms per day. When there is biochemical or radiographic evidence of bone healing, (and in hypoparathyroid patients when normal plasma calcium levels have been attained), the dose generally decreases. Maintenance doses are generally in the range of 0.25 to 1 microgram per day. If hypercalcaemia occurs, Alfacalcidol should be stopped until plasma calcium returns to normal (approximately 1 week) then restarted at half the previous dose.

(a) Renal bone disease

Patients with relatively high initial plasma calcium levels may have autonomous hyperparathyroidism, often unresponsive to Alfacalcidol. Other therapeutic measures may be indicated.

Before and during treatment with Alfacalcidol, phosphate binding agents should be considered to prevent hyperphosphataemia. It is particularly important to make frequent plasma calcium measurements in patients with chronic renal failure because prolonged hypercalcaemia may aggravate the decline of renal function.

(b) Hyperparathyroidism:

In patients with primary or tertiary hyperparathyroidism about to undergo parathyroidectomy, pre-operative treatment with Alfacalcidol for 2-3 weeks alleviates bone pain and myopathy without aggravating pre-operative hypercalcaemia. In order to decrease post-operative hypocalcaemia, Alfacalcidol should be continued until plasma alkaline phosphatase levels fall to normal or hypercalcaemia occurs.

(c) Hypoparathyroidism:

In contrast to the response to parent vitamin D, low plasma calcium levels are restored to normal relatively quickly with Alfacalcidol. Severe hypocalcaemia is corrected more rapidly with higher doses of Alfacalcidol (e.g. 3-5 micrograms) together with calcium supplements.

(d) Neonatal hypocalcaemia:

Although the normal starting dose of Alfacalcidol is 0.05-0.1 microgram/kg/day (followed by careful titration) in severe cases doses of up to 2 microgram/kg/day may be required. Whilst ionised serum calcium levels may provide a guide to response, measurement of plasma alkaline phosphatase activity may be more useful. Levels of alkaline phosphatase approximately 7.5 times above the adult range indicates active disease.

A dose of 0.1 microgram/kg/day of Alfacalcidol has proven effective as prophylaxis against early neonatal hypocalcaemia in premature infants.

(e) Nutritional and malabsorptive rickets and osteomalacia:

Nutritional rickets and osteomalacia can be cured rapidly with Alfacalcidol. Malabsorptive osteomalacia (responding to large doses of IM or IV parent vitamin D) will respond to small doses of Alfacalcidol.

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(f) Pseudo-deficiency (D-dependent) rickets and osteomalacia:

Although large doses of parent vitamin D would be required, effective doses of Alfacalcidol are similar to those required to heal nutritional vitamin D deficiency rickets and osteomalacia.

(g) Hypophosphataemic vitamin D-resistant rickets and osteomalacia:

Neither large doses of parent vitamin D nor phosphate supplements are entirely satisfactory. Treatment with Alfacalcidol at normal dosage rapidly relieves myopathy when present and increases calcium and phosphate retention. Phosphate supplements may also be required in some patients.

Contraindications

Alfacalcidol should not be administered in the presence of hypercalcaemia, hyperphosphataemia (except when occurring with hypoparathyroidism) or hypermagnesaemia. Alfacalcidol should not be used in patients with evidence of Vitamin D toxicity or known hypersensitivity to the effects of Vitamin D or any of its analogues

Warning and Precautions

Alfacalcidol increases the intestinal absorption of calcium and phosphate, serum levels of which should be monitored, particularly in patients with renal failure. Throughout treatment with Alfacalcidol regular plasma and urinary (24- hour collection) calcium levels should be determined at least once every three months months. Alfacalcidol therapy requires regular monitoring of calcium phosphate, alkaline phosphate, magnesium and creatinine levels as well as other appropriate biochemical parameters and should only be prescribed when suitable facilities are available. If there is biochemical evidence of bone healing (eg. return towards normal serum alkaline phosphatase levels), hypercalcaemia may develop if the dose of Alfacalcidol is not decreased appropriately; if hypercalcaemia or hypercalcuria occur, this can be corrected rapidly by stopping treatment with Alfacalcidol and any calcium supplements until plasma calcium levels return to normal, usually in about a week.

Alfacalcidol should be administered with caution to patients with hypercalcaemia especially those with a history of renal calculi.

Drug interactions:

Alfacalcidol/Digitalis/Glycosides

Hypercalcaemia in patients taking digitalis preparations may precipitate cardiac arrhythmias. Patients taking digitalis concurrently with alfacalcidol must therefore be closely monitored.

Alfacalcidol/Barbiturates/Enzyme-inducing Anticonvulsant Drugs

Patients on barbiturates or other enzyme-inducing anticonvulsants may require an increased dose of alfacalcidol to produce the desired effect.

Alfacalcidol/Drugs affecting intestinal absorption

Absorption of alfacalcidol may be impaired by concurrent use of mineral oil (prolonged use), cholestyramine, colestipol, sucralfate or large amounts of aluminium-based antacids.

Alfacalcidol/Magnesium

Caution should be exercised in the use of magnesium-based antacids or laxatives for patients taking alfacalcidol who are on chronic renal dialysis. Hypermagnesaemia may occur.

Alfacalcidol/Calcium/Thiazides

The risk of hypercalcaemia is increased in patients taking calcium-containing preparations or thiazide diuretics concurrently with alfacalcidol.

Alfacalcidol/Vitamin D and Derivatives

Alfacalcidol is a potent derivative of Vitamin D. Pharmacological doses of vitamin D and its derivatives should not be given during alfacalcidol treatment because of the possibility of additive effects and an increased risk of hypercalcaemia.

Pregnancy and Lactation

Pregnancy:

There is insufficient evidence on which to assess the safety of alfacalcidol use during pregnancy. Animal study has not revealed any hazard but as with all drugs Alfacip should only be used during pregnancy if treatment is essential and no better alternative is available.

Lactation

Although not definitely established, it is likely that increased levels of 1,25- dihydroxyvitamin D 3 will be found in the breast milk of mothers treated with alfacalcidol. This might have some influence on calcium metabolism in a breast- feeding infant and discontinuation of breast-feeding should be considered.

Side Effects

Adverse effects generally relate to hypercalcaemia and in the case of renal impairment, hyperphosphataemia which may be induced by alfacalcidol therapy.

Hypercalcaemia may manifest as malaise, fatigue, weakness, dizziness, headache, nausea, dry mouth, constipation, diarrhoea, heartburn, vomiting, abdominal pain or other gastrointestinal discomfort, muscle pain, bone pain, joint pain, pruritus or palpitations. In hypercalcaemic dialysis patients, the possibility of calcium influx from the dialysate should be considered. No other side effects associated directly with alfacalcidol therapy have been noted.

Overdosage:

Overdosage of alfacalcidol can lead to hhpercalcemia which may manifest clinically as malaise, fatigue, weakness, dizziness, drowsiness, headache, nausea, dry mouth,constipation,diarrhea,heartburn,vomiting,abdominal pain or other gastrointestinal discomfort, muscle pain, bone pain, joint pain,pruritis or palpitations. Administration of alfacalcidol should be stopped if hypercalcemia occurs. Severe hypercalcemia may require treatment with general supportive measures, with intravenous fluids, and if needed, with a loop diuretic or corticosteroids.

In acute overdosage, early treatment with gastric lavage and/or the administration of mineral oil may reduce absorption and promote faecal elimination.

Storage

Store in cool, dry & dark place below 30°C.

Shelf life

18 months

Packing/Pack Size

-Carton containing 10 blisters of 10 capsules each

Date of Revision:

July 2022

Cipla

Registration Holder in Malaysia

CIPLA MALAYSIA SDN BHD

(Company No. 1039096-P)

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SAP Code: XXXXXXXX (Ver. 01)

Leaflet size: 150 x 200 mm

Colours:

■ Black

Reference / Supersedes: 21066138

Ph_code / 2D code :

Country : Malaysia

Date: 5-7-2022

Path: F:\Vaibhav\Vinod\Malaysia\

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